Notes on trace formulas for finite groups*

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We will write down some analogs of the "global" (or Selberg) trace formula and the "global" trace formula of Arthur in the case of finite groups. We shall only assume a basic familiarity with groups and representations. For the sake of drawing an analogy, we shall often write sums as integrals.

Notation: Let G be a finite group, G_* the set of conjugacy classes, G^* the set of equivalence classes of irreducible complex representations. We denote a conjugacy class in G by

$$\{\gamma\} = \{g^{-1}\gamma g \mid g \in G\}, \quad \gamma \in G.$$

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We often identify G_* (resp., G^*) with a complete set of representatives of these classes. Let $C(X)$ denote the vector space of complex-valued functions on a finite set X. Let dx denote the counting measure on G.

The vector space $C(G_*)$ is called the space of **class functions** on G.

1 Orbital integrals and characters

We can construct a class function from any function $f \in C(G)$ by summing over all the conjugate elements: Define the **orbital integral** by

$$\begin{aligned} O &: C(G) \to C(G_*), \\ O(f,\gamma) &= \int_G f(x^{-1}\gamma x)\,dx, \quad \gamma \in G. \end{aligned}$$

Here dg denotes the counting measure. For $\pi \in G^*$, let $\mathrm{tr}\ \pi \in C(G_*)$ denote the trace (or **character**) of π. The following two lemmas are well-known (due to Frobenius or Schur):

Lemma 1. *$C(G_*) = \mathrm{span}\{\mathrm{tr}\ \pi \mid \pi \in G^*\}$ and $\dim\ C(G_*) = |G_*| = |G^*|$.*

In other words, the characters form a basis for the vector space of class functions.

This lemma will be referred to as the **completeness lemma**.

Lemma 2. *For $\pi, \pi' \in G^*$, we have*

$$\frac{1}{|G|}\sum_{g\in G} \mathrm{tr}\ \pi(g)\cdot\overline{\mathrm{tr}\ \pi'(g)} = \begin{cases} 1, & \pi \cong \pi', \\ 0, & \text{otherwise.} \end{cases}.$$

This lemma will be referred to as **orthogonality of characters**.

Corollary 3. *For all $\pi \in G^*$, we have*

$$\frac{1}{|G|}\sum_{g\in G} \mathrm{tr}\,\pi(g) = \begin{cases} 1, & \pi \cong 1, \\ 0, & \text{otherwise,} \end{cases} \tag{1}$$

For $f_1, f_2 \in C(G)$, let

$$(f_1, f_2) = \frac{1}{|G|} \sum_{g \in G} f_1(g) \cdot \overline{f_2(g)}.$$

We call this the **Schur inner product** on $C(G)$. Note that the collection of functions

$$\delta_x(y) = \begin{cases} 1, & \text{if } x = y, \\ 0, & \text{if } x \neq y. \end{cases},$$

for $x \in G$, form a basis for the vector space $C(G)$ which is orthonormal with respect to the Schur inner product. The above lemma says that the collection of irreducible characters form an orthonomal basis as well.

Example 4. *Let $G = S_3$, the symmetric group on 3 letters. Let $\sigma = (1\ 2)$, $\tau = (1\ 2\ 3)$, so*

$$G = \{1, \sigma, \tau, \tau^2, \sigma\tau, \sigma\tau^2\}.$$

Using the fact that $\sigma\tau\sigma^{-1} = \tau^2$, it can be shown that G has only 3 distinct conjugacy classes:

$$G_* = \{\{1\}, \{\sigma\}, \{\tau\}\}.$$

In fact,

$$|\{1\}| = 1, \quad |\{\sigma\}| = 3, \quad |\{\tau\}| = 2.$$

There are three inequivalent irreducible complex representations of G:

π_1 : $g \longmapsto$ identity map on $\mathbb{C}$(trivial rep)
π_2 : $g \longmapsto (z \longmapsto sgn(g) \cdot z)$ on $\mathbb{C}$(sign rep)
π_3 : $g \longmapsto$ associated 3×3 permutation matrix (standard rep)
acting on $\mathbb{C}^3/\{(x, y, z) \mid x + y + z = 0\}$.

Their values on the conjugacy classes are summarized as follows:

tr $\pi \backslash \lvert\{\gamma\}\rvert$	*1*	*3*	*2*
tr $\pi \backslash \{\gamma\}$	$\{1\}$	$\{\sigma\}$	$\{\tau\}$
tr π_1	*1*	*1*	*1*
tr π_2	*1*	*-1*	*1*
tr π_3	*2*	*0*	*-1*

We shall explain why the columns of this table are orthogonal in the next section.

Example 5. *The group A_5 of even permutations on $\{1,2,3,4,5\}$ has 5 conjugacy classes:*

$$\{1\}, \quad \{a=(1,2)(3,4)\}, \quad \{b=(1,2,3)\},$$

$$\{c=(1,2,3,4,5)\}, \quad \{d=(1,3,5,2,4)\}.$$

There are 5 ireducible characters. Their values on the conjugacy classes are summarized as follows:

tr $\pi\backslash\lvert\{\gamma\}\rvert$	*1*	*15*	*20*	*12*	*12*
tr $\pi\backslash\{\gamma\}$	$\{1\}$	$\{a\}$	$\{b\}$	$\{c\}$	$\{d\}$
tr π_1	*1*	*1*	*1*	*1*	*1*
tr π_2	*3*	*-1*	*0*	ϕ	$\overline{\phi}$
tr π_3	*3*	*-1*	*0*	$\overline{\phi}$	ϕ
tr π_4	*4*	*0*	*1*	*-1*	*-1*
tr π_5	*5*	*1*	*-1*	*0*	*0*

Here $\phi = (1+\sqrt{5})/2$ *and* $\overline{\phi} = (1-\sqrt{5})/2$.

We close this section with a

Remark 1. *One may show (without using the completeness lemma) that the orbital integral map $O : C(G) \to C(G_*)$ is surjective and that*

$$\ker\ O = \text{span}\{f - f^g \mid f \in C(G),\ \ g \in G\},$$

where $f^g(x) = f(g^{-1}xg)$, for $x, g \in G$. The is equivalent to saying that the sequence

$$0 \to I(G) \xrightarrow{i} C(G) \xrightarrow{O} C(G_*) \to 0,$$

is exact, where $I(G) = C(G)^{G-1} = \{f^g - f \mid f \in C(G),\ g \in G\}$ and i is inclusion.

Since these three groups $I(G)$, $C(G)$ and $C(G_*)$ are all G-module (under the action induced by conjugation), one may ask what their cohomology groups are. Since the action of G on $C(G_*)$ is trivial and since $C(G_*)$ is a finite dimensional complex vector space, we know that $H^1(G, C(G_*)) = Hom(G, C(G_*)) = 0$. What are the other cohomology groups?

2 A Fourier expansion

Let

$$\overline{\pi}(f) = \int_G f(g)\overline{\pi(g)}\, dg = \sum_{g\in G} f(g)\overline{\pi}(g)$$

and

$$\operatorname{tr}\overline{\pi}\,(f) = \int_G \operatorname{tr}\overline{\pi}\,(g) f(g)\, dg = \sum_{g\in G} \operatorname{tr}\overline{\pi}\,(g) f(g).$$

By completeness, each orbital integral may be written as a linear combination of characters. Indeed, we have the following expansion.

Lemma 6. *For $f \in C(G)$ and $\gamma \in G$, we have*

$$O(f,\gamma) = \sum_{\pi\in G^*} \operatorname{tr}\overline{\pi}\,(f)\cdot \operatorname{tr}\pi\,(\gamma). \tag{2}$$

proof: By the completeness lemma, there are constants $a_\pi(f) \in \mathbb{C}$ such that

$$O(f,\gamma) = \sum_{\pi\in G^*} a_\pi(f)\cdot \operatorname{tr}\pi\,(\gamma).$$

Using the orthogonality of characters, we have

$$\begin{aligned}
|G|\cdot a_\pi(f) &= \int_G (\sum_{\pi'\in G^*} a_{\pi'}(f)\cdot \operatorname{tr}\pi'(x))\overline{\operatorname{tr}\pi\,(x)}\, dx \\
&= \int_G O(f,x)\overline{\operatorname{tr}\pi(x)}\, dx \\
&= \int_G\int_G f(g^{-1}xg)\overline{\operatorname{tr}\pi\,(x)}\, dgdx \\
&= \int_G\int_G f(x)\overline{\operatorname{tr}\pi\,(g^{-1}xg)}\, dgdx \\
&= |G|\operatorname{tr}\overline{\pi}\,(f).
\end{aligned}$$

□

For example, let $\gamma' \in G$ and let f denote the characteristic function of the *set* $\{\gamma'\}$. Then $\operatorname{tr}\pi(f) = \operatorname{tr}\pi(\gamma')$, In this case,

$$\sum_{\pi\in G^*} \operatorname{tr}\overline{\pi}(f)\cdot \operatorname{tr}\pi(\gamma) = \sum_{\pi\in G^*} \operatorname{tr}\overline{\pi}(\gamma')\cdot \operatorname{tr}\pi(\gamma).$$

By the lemma above, this is $= 0$ if γ is not conjugate to γ' (and is non-zero if $\gamma \in \{\gamma'\}$). This is equivalent to saying that distinct columns of the character table of a finite group are orthogonal.

3 An analog of the global trace formula

Let Γ be a subgroup of G and let G_γ denote the centralizer of γ in G:

$$G_\gamma = \{g \in G \mid g\gamma = \gamma g\}.$$

Let

$$C(\Gamma\backslash G) = \{\phi : G \to \mathbb{C} \mid \phi(hg) = \phi(g),\ \forall h \in \Gamma,\ \forall g \in G\}.$$

Note that G acts on $C(\Gamma\backslash G)$ by right translation:

$$(R(g)\phi)(x) = \phi(xg^{-1}), \quad x, g \in G.$$

(To check this we must verify that, for each $\phi \in C(\Gamma\backslash G)$, we have (a) $R(1)\phi = \phi$, which is obvious, and (b) $R(g_1)R(g_2)\phi = R(g_1g_2)\phi$, which follows from the equation

$$\begin{aligned}(R(g_1)R(g_2)\phi)(x) &= (R(g_1)\phi_{g_2})(x) = \phi_{g_2}(xg_1^{-1}) \\ &= \phi(xg_2^{-1}g_1^{-1}) = \phi(x(g_1g_2)^{-1}) = (R(g_1g_2)\phi)(x), \quad x, g_1, g_2 \in G,\end{aligned}$$

where $\phi_y(x) = \phi(xy^{-1})$.) In other words, each $g \in G$ gives rise to an automorphism

$$R(g) : C(\Gamma\backslash G) \to C(\Gamma\backslash G).$$

This is called the **right regular representation of G on $C(\Gamma\backslash G)$**. For each fixed $f \in C(G)$, define

$$R(f) : C(\Gamma\backslash G) \to C(\Gamma\backslash G)$$

by

$$(R(f)\phi)(x) = \int_G f(y)(R(y)\phi)(x)\ dy, \quad x \in G.$$

This is called the **right regular representation of** $C(G)$ **on** $C(\Gamma\backslash G)$. We may rewrite this as

$$\begin{aligned}(R(f)\phi)(x) &= \int_G f(y)(R(y)\phi)(x)\ dy \\ &= \int_G f(y)\phi(xy^{-1})\ dy \\ &= \int_G \frac{1}{\text{meas}(\Gamma)} \int_\Gamma f(y)\phi(hxy^{-1})\ dh\ dy \\ &= \int_G K_f(x,y)\phi(y)\ dy,\end{aligned} \tag{3}$$

where

$$K_f(x,y) = \frac{1}{\text{meas}(\Gamma)} \int_\Gamma f(y^{-1}hx)\ dh$$

and where $\text{meas}(\Gamma) = \int_\Gamma 1\ dh$ is the **measure** of Γ. The function $K_f : G \times G \to \mathbb{C}$ is called the **kernel function** of the right regular representation of G on $C(\Gamma\backslash G)$.

Choose the measure dh as the counting measure, so that $\text{meas}(\Gamma) = |\Gamma|$. Another way to write this is

$$K_f(x,y) = \frac{1}{|\Gamma|} \sum_{\gamma\in\Gamma} f(y^{-1}\gamma x).$$

Lemma 7. $\text{tr}(R(f)) = \frac{|\Gamma|}{|G|} \sum_{g\in G} K_f(x,x)$.

proof: The delta functions $\{\delta_x \mid x \in \Gamma\backslash G\}$, form an orthonormal basis for $C(\Gamma\backslash G)$ with respect to the inner product

$$(f_1, f_2) = \frac{|\Gamma|}{|G|} \sum_{x\in\Gamma\backslash G} f_1(x) \cdot \overline{f_2(x)}.$$

These elements may be used to compute a matrix representation of $R(f)$.

The trace is therefore given by

$$\begin{aligned}\mathrm{tr}(R(f)) &= \sum_{g\in\Gamma\backslash G}(R(f)\delta_g,\delta_g) = \frac{|\Gamma|}{|G|}\sum_{g\in\Gamma\backslash G}\sum_{x\in\Gamma\backslash G}(R(f)\delta_g)(x)\delta_g(x)\\ &= \frac{|\Gamma|}{|G|}\sum_{g\in\Gamma\backslash G}\sum_{g\in\Gamma\backslash G}\sum_{y\in G} f(y)(R(y)\delta_g)(x)\delta_g(x)\\ &= \frac{|\Gamma|}{|G|}\sum_{g\in\Gamma\backslash G}\sum_{x\in\Gamma\backslash G}\sum_{y\in G} f(y)\delta_g(xy^{-1})\delta_g(x)\\ &= \frac{|\Gamma|}{|G|}\sum_{g\in\Gamma\backslash G}\sum_{x\in\Gamma\backslash G}\sum_{y\in g^{-1}\Gamma x} f(y)\delta_g(x)\\ &= \frac{|\Gamma|}{|G|}\sum_{g\in\Gamma\backslash G}\sum_{y\in\Gamma} f(g^{-1}hg) = \frac{1}{|G|}\sum_{g\in G}\sum_{y\in\Gamma} f(g^{-1}hg),\end{aligned}$$

as desired. □

Lemma 8. *If*

$$R = \oplus_{\pi\in G^*} m_\pi^\Gamma \pi$$

then

(a) $\mathrm{tr}(R(f)) = \sum_{\pi\in G^*} m_\pi^\Gamma \mathrm{tr}(\pi(f))$, *for all* $f \in C(G)$,

(b) $m_\pi^\Gamma \neq 0$ *if and only if the restriction of* π *to* Γ, $res_\Gamma^G(\pi)$, *contains the trivial representation of* Γ.

proof: We only prove (b). Let $\rho \in G^*$ and $f = \mathrm{tr}(\rho)$. By (a) and orthogonality, $\mathrm{tr}(R(\mathrm{tr}(\rho))) = m_\rho$. By the previous lemma, we have

$$\mathrm{tr}(R(\mathrm{tr}(\rho))) = \frac{1}{|G|}\sum_{g\in G}\sum_{y\in\Gamma}\mathrm{tr}(\rho)(g^{-1}hg) = \sum_{y\in\Gamma}\mathrm{tr}(\rho)(h).$$

If

$$Res_\Gamma^G(\rho) = \oplus_{\sigma\in\Gamma^*} n_\sigma \sigma$$

then, by orthogonality,

$$\sum_{h\in\Gamma}\mathrm{tr}(\rho)(h) = |\Gamma| n_1,$$

where n_1 denotes the multiplicity of the trivial representation of Γ in $res_\Gamma^G(\rho)$. □

Lemma 9. $\int_G K_f(x,x)\,dx = \frac{1}{|\Gamma|}\sum_{\gamma\in\Gamma^G} |G/G_\gamma| O(f,\gamma)$, *where* G_γ *is the centralizer of* γ *and where*

$$\Gamma^G = \{\{g^{-1}hg \mid g \in G\} \mid h \in \Gamma\}$$

is the set of G*-conjugacy classes in* Γ.

This will be called the "geometric side" of the "global" trace formula for finite groups. It is the trace of the operator $R(f)$ in (3).

proof: Since

$$\begin{array}{c} G/G_\gamma \to \{\gamma\} \\ gG_\gamma \longmapsto g\gamma g^{-1}, \end{array}$$

is a bijection, we have

$$\begin{aligned} \int_G K_f(x,x)\,dx &= \frac{1}{|\Gamma|}\sum_{\gamma\in\Gamma}\int_G f(x^{-1}\gamma x)\,dx \\ &= \frac{1}{|\Gamma|}\sum_{\gamma\in\Gamma^G}\sum_{\gamma'\in\{\gamma\}}\int_G f(x^{-1}\gamma' x)\,dx \\ &= \frac{1}{|\Gamma|}\sum_{\gamma\in\Gamma^G} |\{\gamma\}| O(f,\gamma) \\ &= \frac{1}{|\Gamma|}\sum_{\gamma\in\Gamma^G} |G/G_\gamma| O(f,\gamma). \end{aligned}$$

□

On the other hand, substituting the Fourier expansion (2) into the result of the above lemma, we obtain

$$\begin{aligned} \int_G K_f(x,x)\,dx &= \frac{1}{|\Gamma|}\sum_{\gamma\in\Gamma^G} |G/G_\gamma| \sum_{\pi\in G^*} \operatorname{tr}\overline{\pi}(f)\cdot\operatorname{tr}\pi(\gamma) \\ &= \frac{1}{|\Gamma|}\sum_{\pi\in G^*} \operatorname{tr}\overline{\pi}(f) \sum_{\gamma\in\Gamma^G} |\{\gamma\}| \operatorname{tr}\pi(\gamma) \\ &= \frac{1}{|\Gamma|}\sum_{\pi\in G^*} \operatorname{tr}\overline{\pi}(f) \sum_{g\in\Gamma} \operatorname{tr}\pi(g). \end{aligned}$$

This will be called the "spectral side".

Setting the geometric side equal to the spectral side gives the

Theorem 10. *("global" trace formula for finite groups) If R is the right regular representation of $C(G)$ on $C(G/\Gamma)$ then*

$$R = \oplus_{\pi \in G^*} m_\pi^\Gamma \pi$$

then, for all $f \in C(G)$, we have

$$\sum_{\pi \in G^*} m_\pi^\Gamma \mathrm{tr}(\pi)(f) = \sum_{\gamma \in \Gamma^G} |G_\gamma|^{-1} O(f, \gamma) = \frac{1}{|G|} \sum_{\pi \in G^*} \mathrm{tr}\, \overline{\pi}(f) \sum_{h \in \Gamma} \mathrm{tr}\, \pi(h).$$

3.1 Special cases of the "global" trace formula

The following result may be regarded as an analog of the Plancherel theorem.

Corollary 11. *Let $\Gamma = 1$ and let $f \in C(G)$ be arbitrary. Then*

$$f(1) = \frac{1}{|G|} \sum_{\pi \in G^*} (\dim\, \pi) \mathrm{tr}\, \overline{\pi}(f).$$

Example 12. *Assume that Γ is a subgroup of G. If π is an irreducible representation induced from Γ, say $\pi = ind_\Gamma^G \sigma$ with $\sigma \neq 1$, then the Frobenius formula states that*

$$\mathrm{tr}\, \pi(g) = \sum_{x \in G, x^{-1} g x \in \Gamma} \mathrm{tr}\, \sigma(x^{-1} g x).$$

This imples

$$\sum_{g \in \Gamma} \mathrm{tr}\, \pi(g) = \sum_{g \in \Gamma} \sum_{x \in G, x^{-1} g x \in \Gamma} \mathrm{tr}\, \sigma(x^{-1} g x).$$

Orthogonality (1) implies

$$\sum_{g \in \Gamma} \mathrm{tr}\, \sigma(x^{-1} g x) = 0,$$

since $x^{-1} \Gamma x = \Gamma$. Therefore, such induced representations do not contribute to the "trace formula" above.

Consider the case $\Gamma = G$ of the above theorem. Since $\mathrm{tr}\, \overline{1}(f) = \int_G f(g)\, dg$, the orthogonality of characters implies the following result.

Corollary 13. *(a)* $\int_G f(g)dg = \sum_{\gamma \in G_*} |G_\gamma|^{-1} O(f,\gamma)$, *for* $f \in C(G)$.
(b) $\int_G f(g)\, dg = \sum_{\pi \in G^*} m_\pi^G \mathrm{tr}\, \overline{\pi}(f)$.

We call (a) the **Weyl integration formula** for (the finite group) G. (This formula has an easier proof than the one above: as an exercise the reader may try to prove it using the decomposition $G = \bigcup_{\gamma \in G_*} \{\gamma\}$). Part (b) is another form of the Plancherel theorem.

We next consider the case where Γ is a subgroup of G of index 2 in the above theorem. In this case, Γ is automatically a normal subgroup of G. There is a theorem of Clifford which tells us precisely which representations in G^* are induced from Γ.

Lemma 14. *(Clifford) Let* $\pi \in G^*$. *Fix any* $s \in G - \Gamma$, *where* Γ *is a subgroup of* G *of index 2. Either*

(a) the restriction of π *is irreducible, in which case* π *is not induced from a representation of* Γ *but* π *is an irreducible constituent of a representation* $ind_\Gamma^G \sigma$, *for some* $\sigma \in \Gamma^*$ *which satisfies* $\sigma \cong \sigma^s$ *and the restriction of* π *to* Γ *is* σ,

or

(b) the restriction of π *is reducible, say* $\pi = \sigma \oplus \sigma'$ *(some* $\sigma, \sigma' \in \Gamma^*$*) in which case* $\sigma' \cong \sigma^s$, σ' *is inequivalent to* σ, *and*

$$\pi \cong ind_\Gamma^G \sigma \cong ind_\Gamma^G \sigma'.$$

By orthogonality, the sum $\sum_{g \in \Gamma} \mathrm{tr}\, \pi(g)$ vanishes if the restriction of π to Γ is irreducible and non-trivial. But by the remark above, the sum $\sum_{g \in \Gamma} \mathrm{tr}\, \pi(g)$ vanishes if π is induced irreducibly from Γ. By Cliffords' lemma, one of these cases must hold, so we have the following result.

Lemma 15. *Assume* Γ *is a subgroup of* G *of index* 2. *For all* $f \in C(G)$, *we have*

$$\sum_{\gamma \in \Gamma^G} |G/G_\gamma| O(f,\gamma) = \frac{1}{|\Gamma|} \sum_{\pi \in G^*, \pi|_\Gamma = 1} \mathrm{tr}\, \overline{\pi}(f).$$

4 Analog of the local trace formula for finite groups

We try to compute

$$\int_G \int_G f_1(x^{-1}yx) f_2(y)\, dxdy$$

in two ways.

Let

$$h(g) = \int_G f_1(x^{-1}gx) f_2(g)\, dx.$$

The Weyl integration formula gives

$$\begin{aligned}\int_G \int_G f_1(x^{-1}gx) f_2(g)\, dxdg &= \int_G h(g)\, dg \\ &= \sum_{\gamma \in G_*} |G_\gamma|^{-1} O(h, \gamma) \\ &= \sum_{\gamma \in G_*} |G_\gamma|^{-1} O(f_1, \gamma) O(f_2, \gamma).\end{aligned}$$

This will be called the "geometric side".

The Fourier expansion (2) gives

$$\begin{aligned}\int_G \int_G f_1(x^{-1}gx) f_2(g)\, dxdg &= \int_G O(f_1, y) f_2(y)\, dy \\ &= \sum_{\pi \in G^*} \operatorname{tr} \overline{\pi}(f_1) \operatorname{tr} \pi(f_2).\end{aligned}$$

This will be called the "spectral side".

Setting the geometric side equal to the spectral side gives the

Theorem 16. *("local" trace formula for finite groups) For $f_1, f_2 \in C(G)$, we have*

$$\sum_{\pi \in G^*} \operatorname{tr} \overline{\pi}(f_1) \operatorname{tr} \pi(f_2) = \sum_{\gamma \in G_*} |G_\gamma|^{-1} O(f_1, \gamma) O(f_2, \gamma).$$

4.1 Special cases of the local trace formula

Setting $f_1 = f_2$ gives the following curious identity:

Corollary 17. *For $f \in C(G)$ real-valued, we have*

$$\sum_{\pi \in G^*} |\operatorname{tr} \pi(f)|^2 = \sum_{\gamma \in G_*} |G_\gamma|^{-1} O(f, \gamma)^2.$$

Example 18. *If $G = A_5$ and f is the characteristic function of the conjugacy class $\{(1,2,3)\} \subset A_5$ then* [1] $\sum_{\pi \in G^*} |\mathrm{tr}\, \pi(f)|^2 = 400 + 400 + 400 = 1200$ *and* $\sum_{\gamma \in G_*} |G_\gamma|^{-1} O(f,\gamma)^2 = 1200$*, as expected.*

Next, we write the local trace formula for finite groups down more explicitly. First, some notation. If we let $r = |G_*| = |G^*|$ then we can let

$$\begin{aligned} G_* &= \{\gamma_1, ..., \gamma_r\}, \\ G^* &= \{\pi_1, ..., \pi_r\}, \\ G_i &= G_{\gamma_i}, \qquad 1 \leq i \leq r, \\ f_\gamma &= |G|^{-1} ch_\gamma, \qquad \gamma \in G_*, \end{aligned}$$

where ch_X denotes the characteristic function of a finite set $X \subset G$. Finally, let

$$a_{\ell m} = \mathrm{tr}\, \pi_\ell(f_{\gamma_m}), \qquad 1 \leq \ell, m \leq r.$$

(These a_{ij} are equal to $|G_j|^{-1}$ times the ij^{th} entry in the character table for G.) Then $O(\gamma_i, f_j) = \delta_{ij}$, where δ_{ij} is the Kronecker delta function,

$$\delta_{ij} = \begin{cases} 1, & i = j, \\ 0, & i \neq j, \end{cases}$$

and where $a_{ij} = |G_j|^{-1} \mathrm{tr}\, \pi_i(\gamma_j)$.

Let $A = (a_{ij})_{1 \leq i,j \leq r} = (\vec{a}_1, ..., \vec{a}_r)$, where $\vec{a}_k$ is the $k-th$ column vector of the matrix A. If we put $f_1 = f_{\gamma_j}$ and $f_2 = f_{\gamma_i}$ then the trace formula in this case says

$$\langle \vec{a}_i, \vec{a}_j \rangle = |G_i|^{-1} \delta_{ij}, \qquad 1 \leq i, j \leq r,$$

where $\langle \vec{v}, \vec{w} \rangle = \sum_i v_i \overline{w}_i$ is the usual Hermitian inner product on $\mathbb{C}^r$. In other words, the trace formula in this case says

$$A^* A = \mathrm{diag}(|G_1|^{-1}, ..., |G_r|^{-1}),$$

where $A^* = \overline{A^t}$ is the conjugate transpose.

Example 19. *Let $G = S_3$ and let*

$$\gamma_1 = 1, \qquad \gamma_2 = (1\ 2), \quad \gamma_3 = (1\ 2\ 3).$$

[1] This computation was performed with the aid of GAP, a group theory software package. For more details, see [J].

and we label π_1, π_2, and π_3 as before. We have

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = \begin{pmatrix} 1/6 & 1/2 & 1/3 \\ 1/6 & -1/2 & 1/3 \\ 1/3 & 0 & -1/3 \end{pmatrix} = \frac{1}{6} \begin{pmatrix} 1 & 3 & 2 \\ 1 & -3 & 2 \\ 2 & 0 & -2 \end{pmatrix}$$

and

$$|G_1| = 6, \qquad |G_2| = 2, \qquad |G_3| = 3.$$

It is easy to check that

$$A \cdot A^* = \mathrm{diag}(6^{-1}, 2^{-1}, 3^{-1}),$$

as predicted by the trace formula. (However, A is not normal and A^*A is not diagonal.)

Exercise 20. *Compute explicitly both sides of the equation in corollary 17 in case $G = S_3$ and $f = a_1 f_{\gamma_1} + a_2 f_{\gamma_2} + a_3 f_{\gamma_3}$, where a_1, a_2, a_3 are arbitrary real coefficients.*

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